NLP Project Proposal

TempoRAG-KG

Temporal Knowledge Graph-Augmented Retrieval for Multi-Hop Question Answering

Supanut Kompayak · Aphisit Jaemyaem · Dechathon Niamsa-ard · Kaung Hein Htet

Asian Institute of Technology

We noticed a gap in KG²RAG

(Soman et al., 2024)

What the Knowledge Graph stores:

Steve Jobs · CEO_of · Apple

Tim Cook · CEO_of · Apple

⚠ No temporal info — both look equally valid

Query:
"Who was CEO of Apple in 2005?"

System response:

Retrieves both facts
↓
CONFLICT
No way to decide
↓
✗ **Hallucinates**

Key observation: KG triples are stored as timeless facts. the retriever has no mechanism to resolve which is valid at query time.

Why It Matters

35.0%	38.3%	46.7%
HotpotQA questions have temporal expressions	MuSiQue questions have temporal expressions	in 4-hop questions ↑ harder = more temporal

The Research Gap

No existing KG-RAG system — including KG²RAG, GEAR, GraphRAG, or GoG — attaches temporal validity to KG triples **or filters stale facts by time.**

TempoRAG-KG is the first framework to do this on general-domain unstructured text.

Our Solution: TempoRAG-KG

① Temporal Triple Tagging

TempAgent [Xiong et al., NAACL 2025]

Attach [valid_from, valid_to] to every KG triple at construction time — enabling deterministic point-in-time filtering during retrieval.

② GoG Fill-in Mechanism

GoG [He et al., EMNLP 2024]

When temporal filtering leaves dead ends in the graph, invoke LLM parametric knowledge to fill in missing facts (confidence threshold ≥ 0.7).

Temporal Filter (applied before GEAR graph expansion)

Keep edge e iff: (valid_from = null OR valid_from ≤ y_q) AND (valid_to = null OR valid_to ≥ y_q)

Temporal Failure Taxonomy (4 types — all addressed by TempoRAG-KG)

Type 1
Stale Fact

Type 2
Conflicting Facts

Type 3
Missing Context

Type 4
Hop Failure

System Architecture

OFFLINE

Documents → Chunking 512 tok, 100 overlap → LLM Extraction + valid_from / valid_to → Neo4j KG + FAISS

ONLINE

Query → Temporal Detect → Seed Retrieval → Temporal Filter → GEAR Beam Search → GoG Fill-in → MST → LLM → Answer

Our additions

KG^{2}RAG backbone

Evaluation Plan

Datasets

HotpotQA

7,405 questions | 2-hop | 35.0% temporal

MuSiQue

2,417 questions | 2–4 hop | 38.3% temporal

Metrics

F1 & EM (overall + temporal / non-temporal subsets)
Retrieval Precision@5 (gold supporting facts retrieved)
Temporal Extraction F1 (RQ3, ±1 year tolerance)

Ablation Design

Condition	Temporal	GoG	Graph Exp.
Vanilla RAG	✗	✗	✗
KG^{2}RAG	✗	✗	✓
+Temporal only	✓	✗	✓
+GoG only	✗	✓	✓
Full (ours)	✓	✓	✓

Table 2: Ablation conditions. ✓ = component enabled; ✗ = disabled.

Expected Results Table

Model	Overall		Temporal Subset		Non-temporal Subset	
	F1	EM	F1	EM	F1	EM
Vanilla RAG	–	–	–	–	–	–
GraphRAG	–	–	–	–	–	–
KG^{2}RAG	–	–	–	–	–	–
TempoRAG-KG (ours)	**–**	**–**	**–**	**–**	**–**	**–**

Progress & Takeaway

✓ Completed

- Literature review (16 papers)
- EDA: HotpotQA + MuSiQue
- Temporal failure taxonomy (4 types)
- Competitor analysis (EvoReasoner)

▶ In Progress

- Neo4j schema + setup
- Temporal extraction prompts
- GitHub repository setup

☐ Remaining

- KG^2RAG baseline reproduction
- Manual annotation (200 passages)
- Full pipeline implementation
- Ablation experiments
- Streamlit demo

"Temporal validity should be a first-class property of KG edges — not an optional annotation."